

POKER NIGHT

MVP Technical Specification

Implementation-ready requirements for the reduced-cost web-first product

MVP Scope v2.0

Web-first MVP • One administrator • Tournament + minimal cash mode • No Spotify

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1. Document Purpose and Product Boundaries

This document is the implementation source of truth for the Poker Night MVP. It describes what must be built, what must not be built, how users move through the product, how data is stored, and how the tournament calculations behave. When a requirement appears ambiguous, the later and more specific rule in this document takes priority.

Core principle

The administrator chooses the event. Poker Night proposes a playable tournament from the real chips, confirmed attendance and target duration. The administrator remains in control and must confirm all changes that affect play.

1.1 Product objective

- Provide one responsive website/PWA that works on phones, tablets, computers and TV browsers.
- Allow a single administrator to create and operate private No Limit Texas Hold'em tournaments.
- Allow registered players to RSVP, check in, use chat and polls, receive notifications and view the live game.
- Allow guests to check in through a web code without creating an account.
- Generate stacks, blinds, level length, chip composition, ante timing and payouts from practical inputs.
- Provide a separate minimal cash-game tracker.
- Keep the active tournament recoverable on the administrator device if the connection or browser fails.

1.2 Explicit non-goals for this MVP

- No Spotify integration.
- No native iOS or Android applications; the PWA is the launch client.
- No dedicated Chromecast, AirPlay or casting SDK integration. Users may open TV Mode directly or mirror the browser themselves.
- No multiple administrators controlling one game.
- No complex role hierarchy or financial permission system.
- No friends system, public social feed, achievements, advanced seasons or advanced analytics.
- No real-money payment processing and no paid/unpaid tracking.
- No automatic hand tracking, card recognition, dealer rotation or pot calculation.
- No full multi-device offline operation. Only the administrator's active game is locally recoverable.
- No continuous high-complexity tournament simulation. Live recommendations use a lightweight recalculation at level boundaries or on request.

2. Final MVP Scope

Module	Included in MVP	Not included
Platform	Responsive PWA, web authentication, group and game pages	Native mobile apps
Roles	Admin, registered player, guest	Assistant admin, moderator, financial roles
Tournament	Adaptive setup, timer, live controls, seating, rebuys, add-ons, payouts	Hand tracking or automatic rulings
Social	Group chat, polls, in-app/browser notifications	Friends, feed, media gallery, achievements
TV	Read-only web page opened with code/link	Dedicated casting SDKs

Voice	Standard English browser speech announcements	Premium voice packs or voice commands
History	Group game history and basic player aggregate stats	Detailed personal finance history or graphs
Cash game	Fixed blinds, players, buy-ins, top-ups, cash-outs, reconciliation	Advanced cash rules, multiple tables, waiting lists
Offline	Admin local save and restore	Offline player/guest/TV joining and sync

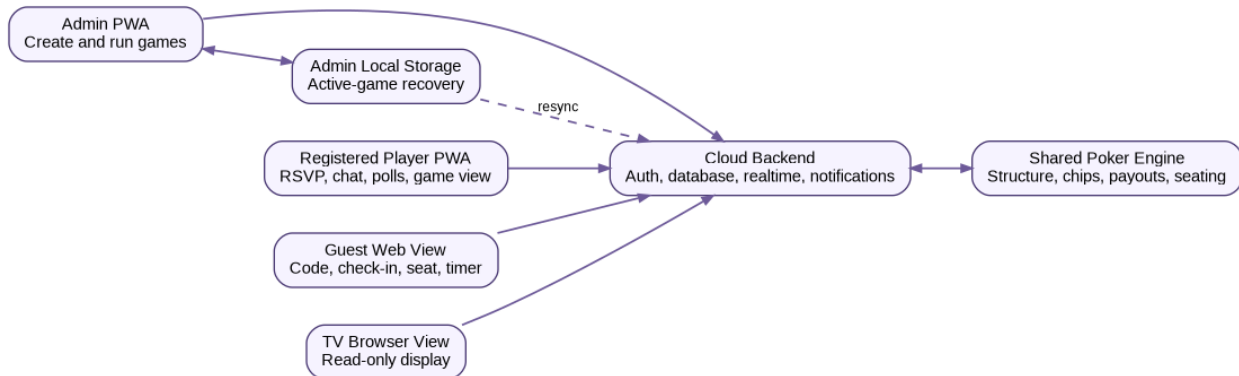


Figure 1 — Reduced-scope system architecture

3. Roles and Permissions

3.1 Administrator

The administrator is a registered user and the only person who can create, edit and operate a game. Each tournament or cash session has exactly one administrator in the MVP.

- Create and edit groups, tournaments, cash sessions, chip sets and presets.
- Publish invitations, create polls and moderate group chat.
- Confirm guests, check in any participant and assign or confirm seats.
- Start, pause, resume and finish the clock.
- Record eliminations, rebuys, re-entries, add-ons and knockouts.
- Accept or reject live structure recommendations and edit future levels.
- See all private calculations, including organizer amount and payout amounts.
- Correct or undo operational actions before the game is locked.

3.2 Registered player

- Join a group with its code or invitation link.
- Respond with Going, Maybe, Can't Come, or Going +1/+2/+3/+4.
- Change RSVP until one hour before the scheduled start.
- Check in and see table/seat after confirmation.
- Use group and tournament chat, answer polls and receive notifications.
- View live public game information, including Prize Pool during the game.
- View completed podium and basic aggregate statistics without money values.

3.3 Guest

- Open the shared tournament web code/link/QR without an account.
- Select the registered inviter and an available guest slot.
- Enter a display name and request check-in.

- Receive administrator confirmation, table and seat.
- View timer, blinds, ante, announcements and seat information.
- Cannot use chat, polls, notifications or permanent group features without registering.

Action	Admin	Registered player	Guest
Create/edit game	Yes	No	No
Operate timer	Yes	No	No
Record rebuy/add-on/elimination	Yes	No	No
RSVP	May edit any	Own response	Guest slot only
Check in	Any participant	Self	Self request
Chat and polls	Yes	Yes	No
View live prize pool	Yes	Yes	Yes
View payout amounts	Yes	No	No

4. System Architecture

4.1 Recommended implementation pattern

The MVP should be a responsive Progressive Web App backed by a managed cloud database with authentication, realtime subscriptions and browser push notifications. Supabase or Firebase are acceptable. The tournament engine must be implemented once as a shared, versioned module and invoked by the server and administrator client with deterministic inputs.

Layer	Responsibility	Suggested technology
Client	Admin, player, guest and TV responsive views	React/Next.js or equivalent PWA
Backend	Auth, validation, data persistence, notifications, realtime fan-out	Supabase/Firebase or small Node backend
Database	Groups, games, participants, structures, actions, chat, polls	PostgreSQL when using Supabase; Firestore otherwise
Realtime	Broadcast authoritative game state to online clients	Database subscriptions or WebSocket channel
Local recovery	Persist administrator game snapshot and queued actions	IndexedDB
Voice	Speak cached announcement text in English	Browser SpeechSynthesis API

4.2 Authoritative state

The cloud record is authoritative while the administrator is online. The administrator client also stores the latest game snapshot and every local action in IndexedDB. If the browser refreshes or temporarily loses connectivity, the clock and controls continue from the local snapshot. Player, guest and TV clients require an online connection and show a stale-state banner if updates stop.

4.3 Timekeeping model

Never decrement and persist the timer every second. Persist timestamps and derive the displayed remaining time.

```
remainingSeconds = max(0, pausedRemainingSeconds - elapsedSinceResume)
```

When running:

```
elapsedSinceResume = serverNow - resumedAt
```

When paused:

```
remainingSeconds = pausedRemainingSeconds
```

Every client corrects local display drift from the latest authoritative server timestamp.

5. Domain Model and Data Ownership

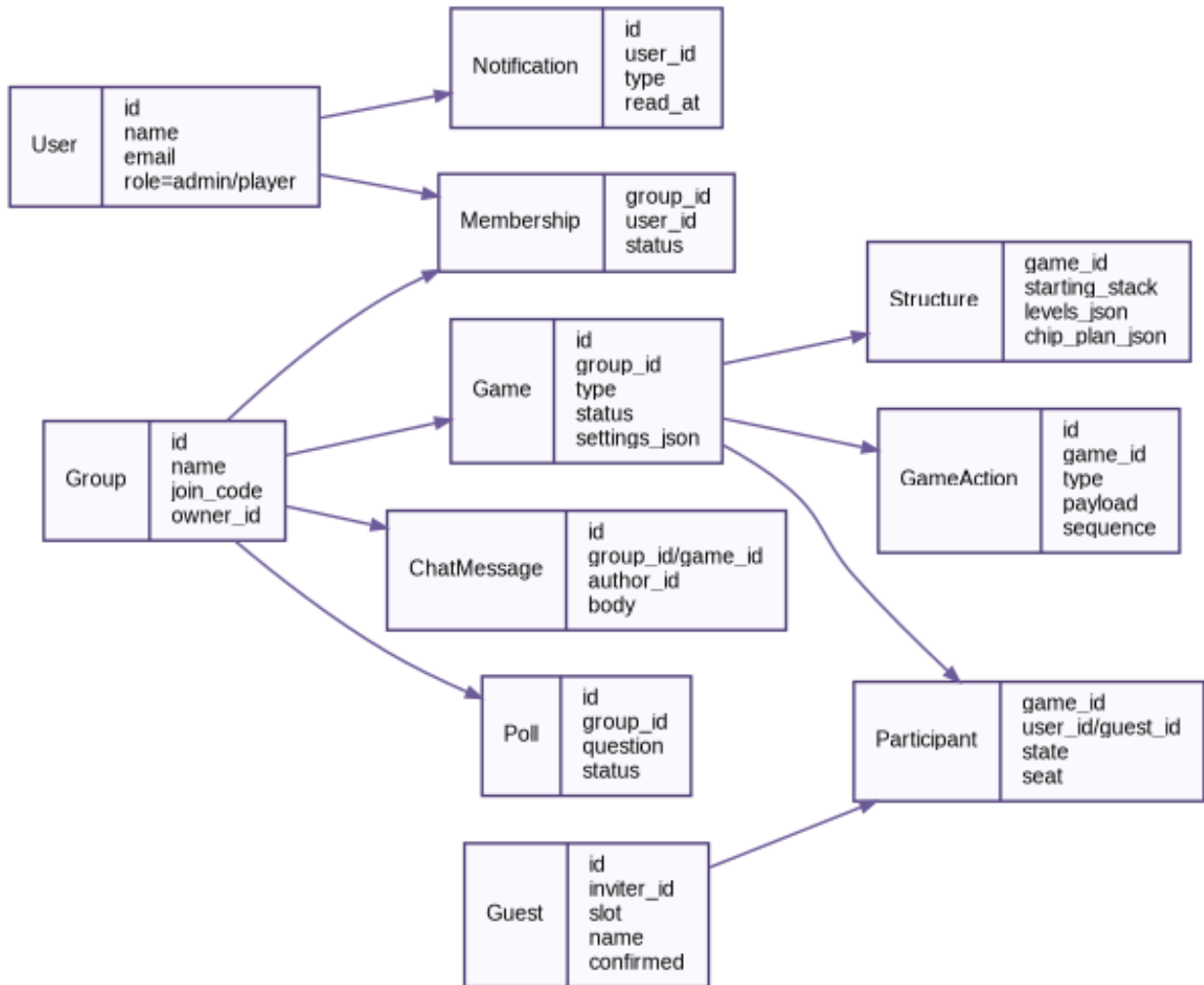


Figure 2 — High-level entity relationships

5.1 Core entities

Entity	Purpose and minimum fields
users	Registered administrators and players. Fields: id, name, email, avatar_url, locale, created_at, status.

groups	Private poker groups. Fields: id, owner_user_id, name, join_code, code_rotated_at, created_at.
group_memberships	User membership in a group. MVP role is admin when owner, otherwise player. Fields: group_id, user_id, status, joined_at.
games	Tournament or cash session header. Fields: id, group_id, admin_user_id, type, name, scheduled_at, location, status, public_code, settings_json, revision.
participants	One person in one game. Supports registered users and guests. Fields: id, game_id, user_id nullable, guest_id nullable, RSVP, check_in_state, table_no, seat_no, status, finish_position.
guests	Temporary person linked to an inviter and slot. Fields: id, game_id, inviter_participant_id, slot_no, name, confirmation_state.
chip_sets	Reusable chip definition. Fields: id, group_id, name, inventory_mode, chips_json.
tournament_structures	Generated proposal and accepted structure. Fields: game_id, engine_version, input_json, starting_stack, levels_json, chip_plan_json, predicted_finish_range.
game_actions	Append-only audit log. Fields: id, game_id, sequence, actor_user_id, type, payload_json, created_at, reversed_by_action_id.
chat_messages	Group or game message. Fields: id, scope_type, scope_id, author_user_id, body, created_at, deleted_at.
polls / poll_options / poll_votes	Simple single-choice or multi-choice group polls.
notifications	User notification inbox and browser-push state.
cash_transactions	Chip-value records only: BUY_IN, TOP_UP or CASH_OUT. They do not represent payment status.
game_results	Completed podium, paid places privately, knockout count and summary statistics.

5.2 Data visibility projections

Never send the same full game object to every role. Build explicit projections: admin projection, registered-player projection, guest projection and TV projection. Private fields such as organizer percentage, organizer amount, payout amounts, rebuy/add-on identities and cash reconciliation must be omitted at the API level, not merely hidden in the interface.

6. Tournament Setup Engine



Figure 3 — Complete tournament user flow

6.1 Required setup inputs

Field	Type and validation	Default
Tournament name	1–80 characters	Poker Night + date
Date/time	Future local date/time	Required
Location	0–160 characters	Optional
Expected players	Integer ≥ 2 , from RSVP or admin override	Going responses + guests
Target duration	3, 3.5, 4, 4.5, 5, 5.5 or 6 hours	3.5 hours
Buy-in	Positive number; no currency symbol in game UI	Required
KO bounty	Off or positive amount displayed as buy-in + bounty	Off
Rebuys	Enabled/disabled, closing level, unlimited or limit	On, end of Level 6, unlimited
Add-on	Enabled/disabled and price only	On, price = buy-in, max 1/player
Ante	Off, big blind ante, individual ante	Off; big blind ante recommended if enabled
Organizer percentage	0–100, private	Group preset or 0
Chip set	Saved, exact inventory or quick inventory	Required

6.2 Preset matching

Before starting from zero, the system searches the administrator's presets. A matching preset is suggested when buy-in, KO setting, target duration, expected attendance, rebuy/add-on settings and chip set are sufficiently close. If two presets match, show both and explain the differences. A preset supplies settings only; the engine always regenerates the structure using current attendance and chips.

6.3 Generation workflow

1. Normalize chip denominations and verify that no two colours share a value.
2. Estimate legal opening blind candidates from the smallest practical denominations.
3. For each candidate, calculate a starting stack and target big-blind depth appropriate to duration and field size.
4. Estimate total chips after expected rebuys and add-ons.
5. Choose one level duration from 10, 15 or 20 minutes as the default for the whole event.
6. Calculate the approximate number of levels available inside the target playing duration.
7. Generate a smooth big-blind curve and snap every level to a chip-compatible amount.
8. Select a simple payable small blind, usually 40–50% of the big blind but not necessarily exactly half.
9. Add the approved ante from the configured level and recalculate pressure.
10. Generate starting, rebuy, add-on and colour-up chip plans.
11. Calculate expected finish range and payout proposal.
12. Return proposal, warnings and alternatives for administrator confirmation.

7. Chip Inventory and Stack Generation

7.1 Inventory modes

Mode	Admin input	Guarantee
Saved	Select an existing exact or quick chip set	Same guarantee as saved mode
Exact	Colour, value and exact quantity for each chip	Engine must not exceed inventory
Quick numbered	Colours, printed values, rank from most to least available	Provisional; admin confirms availability
Quick unnumbered	Colours and availability rank; engine proposes unique values	Provisional; admin confirms mapping and quantities

7.2 Stack objectives

- Starting stack value and opening blinds must be solved together; there is no unconditional 5,000 minimum.
- Provide useful early depth without creating unnecessarily huge physical stacks.
- Use enough low chips to post early blinds without constant change.
- Reserve inventory for expected rebuys, one add-on per player and later colour-ups.
- Prefer visually countable groups and few denominations per stack.
- Never give two chip colours the same value.

7.3 Constructible stack search

```

For each candidate total stack S:
  enumerate practical combinations of active denominations
  reject combinations that exceed per-colour inventory reserve
  score =
    early_blind_payability
  + counting_simplicity
  + stack_aesthetics
  + rebuy_reserve_health
  + colour_up_efficiency
  - excessive_chip_count
Choose the highest-scoring valid combination.

```

7.4 Rebuy stack

A rebuy is available only after elimination and has exactly the same total value as the original starting stack. Its physical composition may change. At later levels, use fewer obsolete small chips and more medium/high chips while keeping enough chips to post the current blinds easily.

7.5 Chip exchanges

The system provides direct instructions such as “exchange 20 white chips for 2 blue chips.” Exact exchanges are preferred. When a small remainder cannot be exchanged exactly, the MVP default is to round the player upward and add the resulting small increase to total chips in play. A formal chip race is an advanced option, not the default.

8. Blind, Level and Ante Generation

8.1 Level length

Every level is exactly 10, 15 or 20 minutes. The generated structure should normally use one duration throughout. During play the administrator may accept a recommendation that changes future levels to another allowed duration. The active level is never changed.

8.2 Starting depth model

```
targetStartingBB = clamp(
    125
    + 28 * (targetHours - 3.5)
    - 2.5 * max(0, expectedPlayers - 8),
    80,
    240
)

candidateStartingStack = practicalRound(openingBB * targetStartingBB)
```

The constants are configuration values, not UI settings, and must be calibrated with test simulations.

8.3 Final blind target

```
expectedTotalChips = startingStack * expectedPlayers
                    + startingStack * expectedRebuyCount
                    + addOnStack * expectedAddOnCount

targetFinalBB = expectedTotalChips / (2 * targetHeadsUpAverageBB)
where targetHeadsUpAverageBB defaults to 15.
```

8.4 Blind curve

```
rawBB(level i) = openingBB * growthFactor^i

growthFactor = (targetFinalBB / openingBB)^(1 / max(1, plannedLevels - 1))

For each raw value:
    snap to a legal, easy-to-post amount
    enforce monotonic increase
    avoid unreasonable jumps
    choose a practical small blind near 40-50% of BB
    allow values such as 20/50 or 1200/2500 when easier with the chips.
```

8.5 Ante

- Default when enabled: big blind ante equal to the current big blind.
- Default activation: after the rebuy period, normally the first level after Level 6.
- Big blind ante remains active short-handed and heads-up by default.
- Individual ante candidate = big blind divided by expected table size, snapped to a practical chip value.
- Enabling an ante triggers a full remaining-duration recalculation before confirmation.

8.6 Live speed-up and slow-down

At a level boundary or when the administrator requests recalculation, compare actual players remaining with the expected curve and estimate the new finish. When the difference exceeds 20 minutes, offer a recommendation. Nothing changes automatically.

Action	Permitted changes	Restriction
--------	-------------------	-------------

Speed up	20→15, 15→10, activate approved ante, increase future blinds	Starts next level only
Slow down	10→15, 15→20, insert intermediate future level, soften growth, delay ante	Never remove a level
Manual edit	Edit a future blind or duration and preview finish impact	Cannot edit completed/active level

9. Prize and Organizer Calculations

9.1 Contribution rules

- Regular buy-in component, rebuys, re-entries and add-ons contribute to the calculation.
- KO bounty money is excluded. In 15 + 5, only 15 contributes to the regular prize pool.
- Poker Night records game contributions for calculation but does not record whether cash was paid.
- During play, players may see the total Estimated Prize Pool and then Prize Pool after settlement.
- After the game, players see paid positions/podium but no money amounts.

9.2 Organizer amount rounding

```
grossEligible = sum(regular entry + rebuy + re-entry + add-on components)
targetOrganizer = grossEligible * organizerPercentage
```

Select organizerAmount nearest to targetOrganizer such that:

```
prizePool = grossEligible - organizerAmount
prizePool is divisible by 10
```

If equally near, choose the lower organizerAmount.

Example: gross 165, target 16.5 -> choose 15, prize pool 150.

9.3 Paid places

The number of paid places is a combination of unique players and prize pool. The engine should generally pay around 15–25% of unique players, subject to a meaningful lowest prize. It may pay two places for small fields/pools, three for medium games and four or more for larger games. The administrator may override.

9.4 Payout weights and rounding

Create descending raw weights:

```
weight_i = exp(-lambda * i), i starting at 0
```

Normalize weights to the prize pool.

Use lambda calibrated to approximate:

```
2 places: ~73/27
```

```
3 places: ~57/30/13
```

```
4 places: ~56/30/10/4
```

Round every displayed payout to a multiple of 10.

Protect the lowest paid place, then second/third balance.

First place absorbs the final 10-unit reconciliation.

Never generate a payout ending in 5.

10. Invitations, Guests and Check-in

10.1 RSVP choices

- Going
- Maybe

- Can't Come
- Going +1
- Going +2
- Going +3
- Going +4

The response is one combined option. Going +2 means the registered player plus two guests, for three confirmed people. RSVP may be changed until one hour before the scheduled start.

10.2 Guest web flow

1. Guest opens the tournament code, link or QR page.
2. Guest chooses “Join as guest.”
3. Guest selects the registered person who invited them.
4. Guest selects one available guest slot attached to that inviter.
5. Guest enters a name and submits check-in.
6. The slot becomes reserved; duplicate selection is blocked.
7. Administrator confirms or rejects the guest.
8. After confirmation, guest receives table and seat and limited live access.

10.3 Registration closing

Late registration closes permanently when the rebuy level ends, normally after Level 6. No player or guest can be added after that point. A late player added before closing receives a fresh full starting stack and a system-recommended table/seat that the administrator confirms.

11. Live Tournament Operations

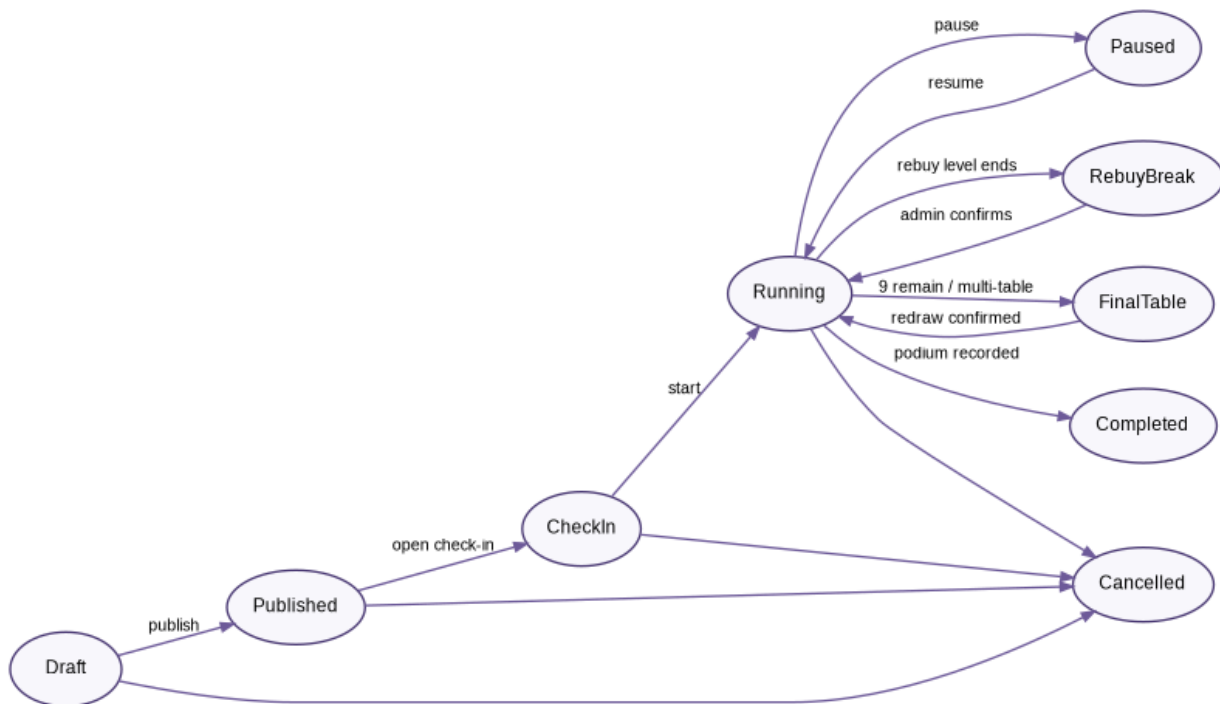


Figure 4 — Tournament state machine

11.1 Admin control screen

- Large clock with Start, Pause, Resume and Next Level.
- Current blinds/ante, next level and current player count.
- Participant list with Check in, Eliminate, Rebuy, Add-on and Undo actions.
- Tables and seats with pending move recommendations.
- Estimated finish and Recalculate button.
- Speed Up, Slow Down and Edit Future Structure buttons.
- Announcements panel and TV link/code.
- Persistent offline/recovery indicator.

11.2 Rebuy-period closing flow

1. When the last rebuy level reaches zero, the clock pauses automatically.
2. Rebuys and late registration close.
3. No fixed break countdown is used.
4. Show final eligible rebuy actions, recommended add-on value, add-on selection for each active player, chip exchanges and ante confirmation.
5. Administrator records final rebuys and up to one add-on per active player.
6. Recalculate total chips, prize pool and future finish estimate.
7. Administrator confirms settlement and manually starts the next level.

11.3 Elimination and undo

Only the administrator records an elimination. The action marks the player eliminated, captures the position and optional single knockout recipient, and updates remaining players. Before another dependent action makes reversal unsafe, the administrator can Undo. Otherwise use Correct Result, which appends a compensating audit action; never delete audit history.

11.4 Lightweight live recommendations

At every level end, calculate actual remaining players, average stack, average big blinds and estimated finish. If estimated finish differs from target by more than 20 minutes, show a non-blocking recommendation. The administrator may ignore it. The app never silently changes blinds, duration or ante.

12. Seating and Final-table Logic

12.1 Initial seating

The administrator chooses Fully Random, Manual, Keep Guests with Inviter, or Separate Guests from Inviter. Default table capacity is nine. Ten or more checked-in participants create multiple tables automatically.

12.2 Table balancing

When tables differ by more than one active player, recommend a move. The administrator sees the source player, target table and target seat, physically completes the move, then confirms. Only after confirmation does the system update the seat map.

12.3 Final table

When nine players remain and the game previously used multiple tables, pause and create a complete random redraw for all nine seats plus a random initial dealer-button position. The administrator confirms seating, then resumes. If the game began on one table, no redraw is required. The app does not track dealer movement afterward.

13. TV Mode and Voice Announcements

13.1 TV Mode

TV Mode is a read-only web page opened with the tournament code or direct link. It may be opened on several displays. The MVP does not implement casting SDKs; users can open it on a TV browser or mirror the page manually.

- Tournament name and state
- Large timer
- Current blinds and ante
- Next level
- Players remaining
- Average stack and average BB
- Estimated Prize Pool before settlement and Prize Pool afterward
- Break/settlement messages
- Visual announcements
- Final-table redraw and completed podium

Never show organizer percentage, organizer amount, individual payouts, rebuy/add-on counts, transaction identities, admin controls or private statistics.

13.2 Voice

The administrator manually selects the one browser/device that plays announcements. Use the browser's standard English speech voice. Other connected screens remain silent.

- Tournament starts
- New level
- Five minutes remaining
- One minute remaining
- Rebuys closed
- Add-ons available
- Final table
- Initial/final-table dealer assignment
- Winner

Elimination names are optional per tournament and disabled by default. Voice commands are deferred.

14. Chat, Polls and Notifications

14.1 Chat

- One group chat and optional tournament-specific chat thread.
- Registered members only; guests cannot read or write chat.
- Plain text only in the MVP, maximum 1,000 characters.
- Admin can delete a message; deleted messages remain in audit metadata but are hidden from users.
- Realtime delivery, unread count and basic spam rate limit.

14.2 Polls

- Admin creates single-choice or multi-choice polls with 2–10 options.
- Use for date, time, buy-in, KO setting, duration, location or custom questions.
- Registered members vote; guests cannot vote.
- Admin closes a poll manually. Results may suggest a matching tournament preset.

14.3 Notifications

Provide in-app notification inbox and browser push when permission is granted. If push is unavailable or refused, the in-app inbox is the fallback. Email and SMS are not required.

Trigger	Recipients	Example
Tournament published	Group members	Friday Poker is open for RSVP
RSVP deadline approaching	No response / Maybe players	RSVP closes in one hour
Guest confirmed	Inviter and guest while page open	Guest 1 is confirmed
Seat assigned	Checked-in participant	Table B, Seat 4
Poll created/closing	Registered members	Vote on the buy-in
Game starting	Going participants	Tournament starts soon
Important admin announcement	Active participants	Rebuy period ends after this level

15. Game History and Basic Statistics

The MVP keeps a group-level history of completed tournaments and cash sessions. It does not provide a detailed personal game ledger or financial history.

15.1 Completed game history

- Date, name, type and attendance
- Podium/all paid positions without payout amounts for players
- Duration and final level
- Basic setup summary
- Admin-only private calculation summary

15.2 Basic player aggregate statistics

- Games played
- Wins
- Podium finishes
- Average finish
- Knockouts

No ROI, profit, investment, winnings, rebuy/add-on history, graphs, streaks or advanced filters. Public/player views contain no money amounts after completion.

16. Minimal Cash-game Module



Figure 5 — Minimal cash-game flow

16.1 Included

- Create a cash session with name, date, location, fixed small blind, fixed big blind and optional buy-in limits.
- Add registered players or guests and assign seats.

- Start a session-duration counter; there are no levels or blind increases.
- Admin records each player's initial buy-in chip value and later top-ups.
- Admin records final cash-out chip value when the player leaves or at session end.
- Reconcile total bought chip value against total cashed-out chip value.
- Save a private admin session summary and a non-financial group history item.

16.2 Excluded

- No waiting list, multiple cash tables, straddles, bomb pots, run-it-twice controls or cash voice announcements.
- No long-term cash profit/loss statistics for players.
- No payment tracking. Values represent chips issued and returned, not confirmation that money changed hands.

16.3 Reconciliation

```
totalIssued = sum(initial buy-ins + top-ups)
totalReturned = sum(cash-outs)
difference = totalReturned - totalIssued
```

```
If difference != 0:
    block final confirmation until admin either corrects records
    or saves an explicit unresolved-difference note.
```

17. Screen-by-screen Specification

Screen	Required behaviour
Welcome / authentication	Sign in, register, enter group code, enter tournament code. Preserve intended destination after authentication.
Home	Upcoming games, groups, notifications, create button for admin, current active game shortcut.
Group	Upcoming/past games, members, chat, polls, presets, group code.
Create tournament	Preset suggestion, basic inputs, chip set, optional rules, generate button.
Structure review	Starting stack, chip breakdown, level table, ante, predicted finish, warnings, Regenerate, Edit, Confirm.
Invitation / RSVP	Single RSVP option, guest count embedded, deadline, share code/link/QR.
Check-in	Going/guest list, pending guest requests, confirm, assign seating.
Admin live dashboard	Clock, levels, participant actions, settlement, structure change, seating, announcements, recovery state.
Player live view	Timer, blinds, next level, players remaining, average stack, prize pool during game, seat, announcements.
Guest live view	Same limited live information plus own seat; no chat/polls/notifications.
TV Mode	Large read-only display, no navigation or controls.
Rebuy/add-on settlement	Final rebuys, add-on value, one checkbox per active player,

	exchanges, ante, prize pool, Confirm and Continue.
Final-table redraw	Nine seats, dealer position, Confirm Seating.
Complete tournament	Enter finishing order, verify podium, admin private payout summary, publish result.
Cash session	Fixed blinds, seated players, timer, buy-in/top-up/cash-out controls, reconciliation.
History / stats	Group history and five basic aggregate statistics.

18. API Contract

The exact transport can be REST or framework server actions, but these logical operations and role checks are required. Use idempotency keys for game actions that could be double-submitted.

Endpoint	Purpose	Permission
POST /auth/register	Create account	Public
POST /groups	Create group and join code	Authenticated admin
POST /groups/{id}/join	Join by code	Authenticated player
GET /groups/{id}	Role-specific group projection	Member
POST /groups/{id}/chat	Send message	Registered member
POST /groups/{id}/polls	Create poll	Admin
POST /polls/{id}/votes	Vote	Registered member
POST /games	Create tournament/cash session	Admin
PATCH /games/{id}	Edit draft/published fields	Admin
POST /games/{id}/generate	Generate tournament proposal	Admin
POST /games/{id}/publish	Publish invitation	Admin
POST /games/{id}/rsvp	Set own RSVP	Registered player
POST /public/games/{code}/guests	Reserve guest slot	Public guest
POST /games/{id}/participants/{pid}/confirm	Confirm guest/check-in	Admin
POST /games/{id}/actions	Append operational action	Admin
POST /games/{id}/structure/recalculate	Preview future changes	Admin
POST /games/{id}/structure/accept	Accept future structure revision	Admin
GET /public/games/{code}/tv	TV projection	Public with code
GET /public/games/{code}/guest	Guest projection	Guest session

POST /games/{id}/complete	Validate and complete	Admin
GET /groups/{id}/history	Group history	Member

18.1 Game action payload

```
{
  "idempotencyKey": "uuid",
  "expectedRevision": 42,
  "type": "ELIMINATE_PLAYER",
  "payload": {
    "participantId": "...",
    "knockoutParticipantId": "... optional"
  },
  "clientOccurredAt": "ISO-8601"
}
```

Reject with HTTP 409 when expectedRevision does not match. Because only one admin exists, conflicts should be rare; nevertheless, browser retries and restored offline actions can cause duplicates.

19. Realtime Event Contract

Event	Payload purpose
game.snapshot	Complete role-safe state after join/reconnect
clock.changed	start, pause, resume, level transition
participant.changed	RSVP, check-in, seat, elimination
structure.changed	accepted future structure revision
prize_pool.changed	estimated/confirmed total only for public roles
announcement.created	visual text and optional speak flag
chat.created	new registered-member message
poll.changed	poll opened, closed or vote totals
notification.created	new inbox item
game.completed	final non-financial result projection

Every event includes game/group ID, monotonically increasing revision/sequence, server timestamp and a projection safe for the recipient role. On missed sequence, the client requests a full snapshot.

20. Offline Recovery and Conflict Rules

20.1 Local persistence

- Write the current admin game snapshot to IndexedDB after every accepted action.
- Store the running clock as timestamps, not a second-by-second counter.
- Store unsynchronised actions with idempotency keys and expected revision.
- On reopen, offer “Restore active tournament” with last saved local time and state.
- When connection returns, submit queued actions in order and stop on first conflict.

20.2 Online viewers

Player, guest and TV views show the last received state plus a visible “Connection interrupted” banner. The timer may continue visually from the last trusted timestamp, but other values are frozen until reconnection. New devices cannot join while the backend is unavailable.

20.3 Recovery safety

If the server game advanced after the local snapshot, do not overwrite it. Show a comparison and require the administrator to choose the server version or manually reapply missing actions. Never merge two different clock states automatically.

21. Validation, Errors and Edge Cases

Scenario	Required behaviour
No legal blind with chips	Reject proposal and show which denomination is missing or propose new values for unnumbered chips.
Quick inventory shortage	Show required quantities; let admin choose fewer players, another stack, another chip mapping or confirm more chips exist.
Duplicate guest slot	First reservation wins temporarily; second guest must choose another slot.
Guest no-show	Admin removes confirmation before start; seating regenerates or seat remains empty until confirmed.
RSVP changed after deadline	Block player change; admin may edit before late registration closes.
Player added during play	Allowed only before rebuy period closes; full fresh stack; recalculate and offer future changes.
Rebuy requested for active player	Block; rebuy only after elimination.
Second add-on	Block; maximum one add-on per active player.
Elimination mistake	Undo if safe, otherwise compensating correction action.
Level reaches zero while offline	Admin client transitions locally and queues the transition.
Browser sleeps	Wake from timestamps; do not add sleep duration twice.
TV opens old code	Show invalid/expired code; never expose prior private data.
Prize pool not divisible by 10	Adjust private organizer amount according to nearest valid rule.
Payout rounding mismatch	First place absorbs final 10-unit reconciliation.
Nine-player final table trigger repeated	Use idempotent final-table state; create only one redraw.
Cash reconciliation mismatch	Require correction or explicit unresolved note before completion.
Notification permission denied	Use in-app notification inbox only.
Chat abuse	Rate limit, admin delete, sanitize text and block markup/scripts.

22. Security and Privacy

- Use managed authentication, secure password reset and verified email where supported.
- All authorization is enforced server-side. Never trust a role or game ID supplied only by the client.
- Tournament codes are random, case-insensitive and exclude ambiguous characters. Rate-limit code lookup.
- Guest sessions use scoped, expiring tokens tied to one guest record and one tournament.
- Sanitize all chat, poll and name input. Store plain text in MVP.
- Do not expose organizer amount, payout amounts or contribution identities in player/guest/TV responses.
- Use HTTPS, secure cookies/token storage, CSRF protections where applicable and database row-level security.
- Keep append-only game actions for audit and restore. Corrections create new actions.
- Provide account deletion and group member removal. Retain anonymized completed-game integrity where legally appropriate.
- Apply rate limits to login, code lookup, guest reservation, chat and voting.

23. Testing and Acceptance Criteria

23.1 Engine tests

- Property test: every generated blind and ante is payable with active chip denominations.
- Property test: starting and rebuy stack totals equal their declared value.
- Exact-inventory test: total allocated chips never exceed configured quantities.
- Property test: blind sequence is strictly increasing and contains no removed level after an accepted revision.
- Property test: level duration is always 10, 15 or 20.
- Property test: payouts total exactly the Prize Pool and all payouts are multiples of 10.
- Property test: KO bounty money never enters regular prize pool or organizer calculation.
- Simulation test: generated finish estimate falls within a reasonable calibrated range for representative fields.

23.2 User-flow acceptance scenarios

Scenario	Pass condition
Create and run standard tournament	Admin creates 8-player, 3.5-hour tournament, confirms generated structure, publishes, checks in players, runs timer, closes rebuys, records add-ons, finishes and publishes podium.
Guest check-in	Player RSVP is Going +2; both guests use code, choose distinct slots, admin confirms, seats appear.
Live speed adjustment	At level end projected finish is 30 minutes late; admin previews Speed Up, accepts 15→10 starting next level, TV and player views update.
Offline recovery	Admin loses connection, pauses and records elimination locally, reloads browser, restores game and later synchronizes without duplicate actions.
TV privacy	TV code shows live game but response payload contains no organizer, payout or participant contribution data.
Cash reconciliation	Admin records two buy-ins, one top-up and two cash-outs; completion succeeds only when totals reconcile or note is saved.

23.3 Definition of done

- All required screens work responsively at common phone, tablet, desktop and TV widths.

- Core flows pass automated and manual acceptance tests.
- No private financial fields appear in player, guest or TV network responses.
- Admin can restore an active game after browser refresh and temporary connection loss.
- Production deployment, environment configuration, database migrations and seed/demo data are documented.
- Source code, setup instructions and basic architecture notes are delivered.

24. Deployment and Operations

- Separate development and production environments.
- Environment variables for backend URL, auth keys, push credentials and error monitoring.
- Automated database migrations or documented schema deployment.
- HTTPS production domain and PWA manifest/service worker.
- Basic error monitoring and structured server logs without sensitive payloads.
- Daily managed database backups when available.
- Seed group and demo tournament for acceptance testing.

25. Deferred Features and Future Phases

Deferred feature	Reason
Native iOS/Android apps	Duplicate client cost; PWA proves product first.
Spotify	Removed from scope.
Dedicated casting SDKs	TV browser/link is sufficient for MVP.
Multiple administrators	Requires conflict resolution and complex permissions.
Advanced tournament health simulation	MVP uses lightweight recalculation.
Premium voices and voice commands	Standard English browser voice is sufficient.
Friends, feed, achievements, seasons	Not core to tournament operation.
Advanced player financial statistics	Privacy and complexity; explicitly excluded.
Full offline multi-device sync	High conflict and networking complexity.
Advanced cash-game features	Cash mode is intentionally minimal.
Payment processing	Regulatory and operational complexity.